Lab4

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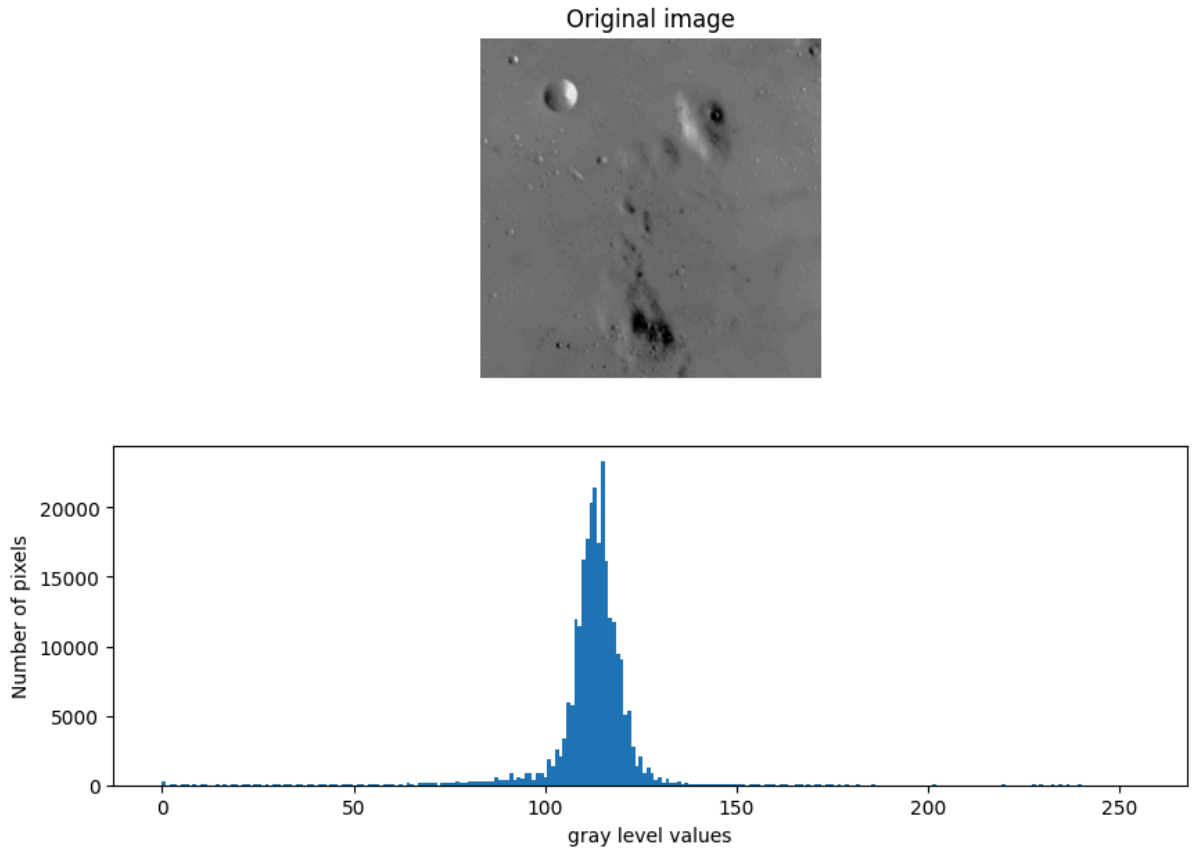
Group: 7MA1

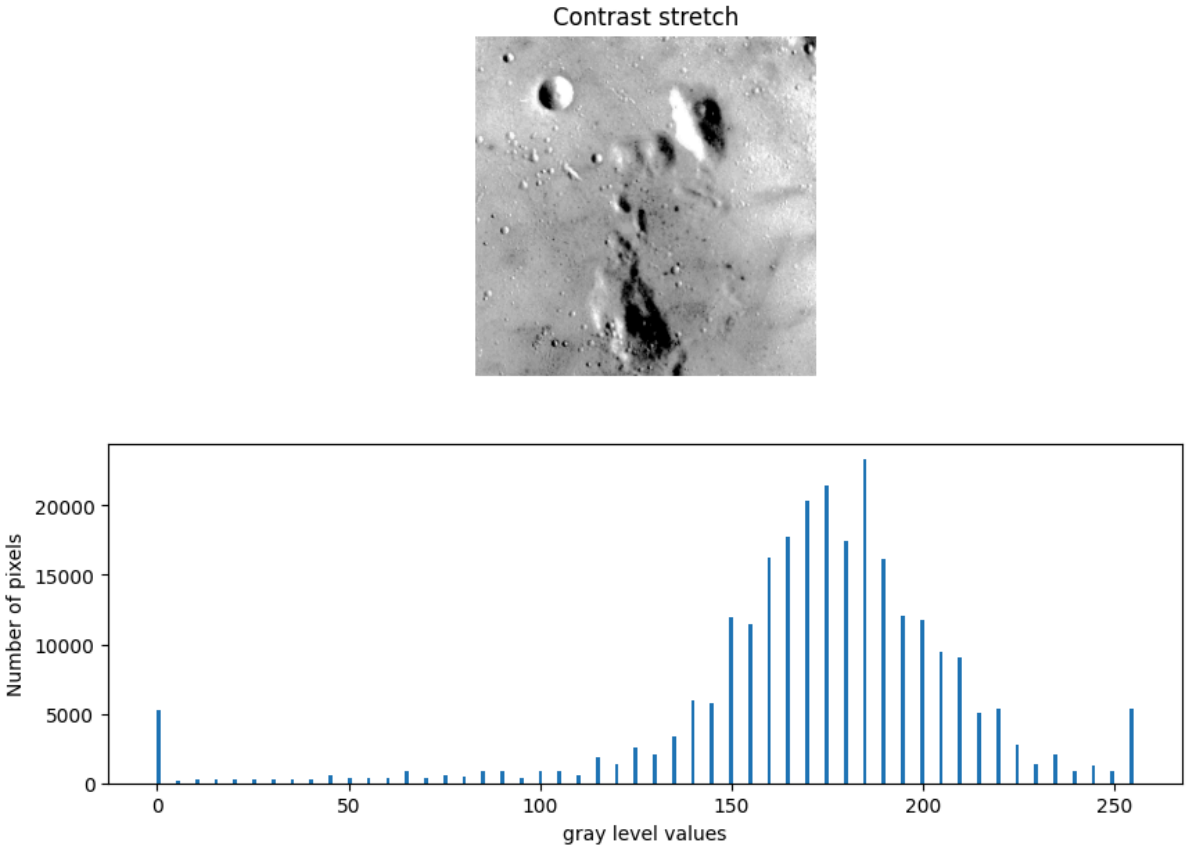
In [2]:
```
import matplotlib.pyplot as plt
import numpy as np
from skimage import data, img_as_float
from skimage import exposure
```

In [ ]:
```
img = data.moon()
```

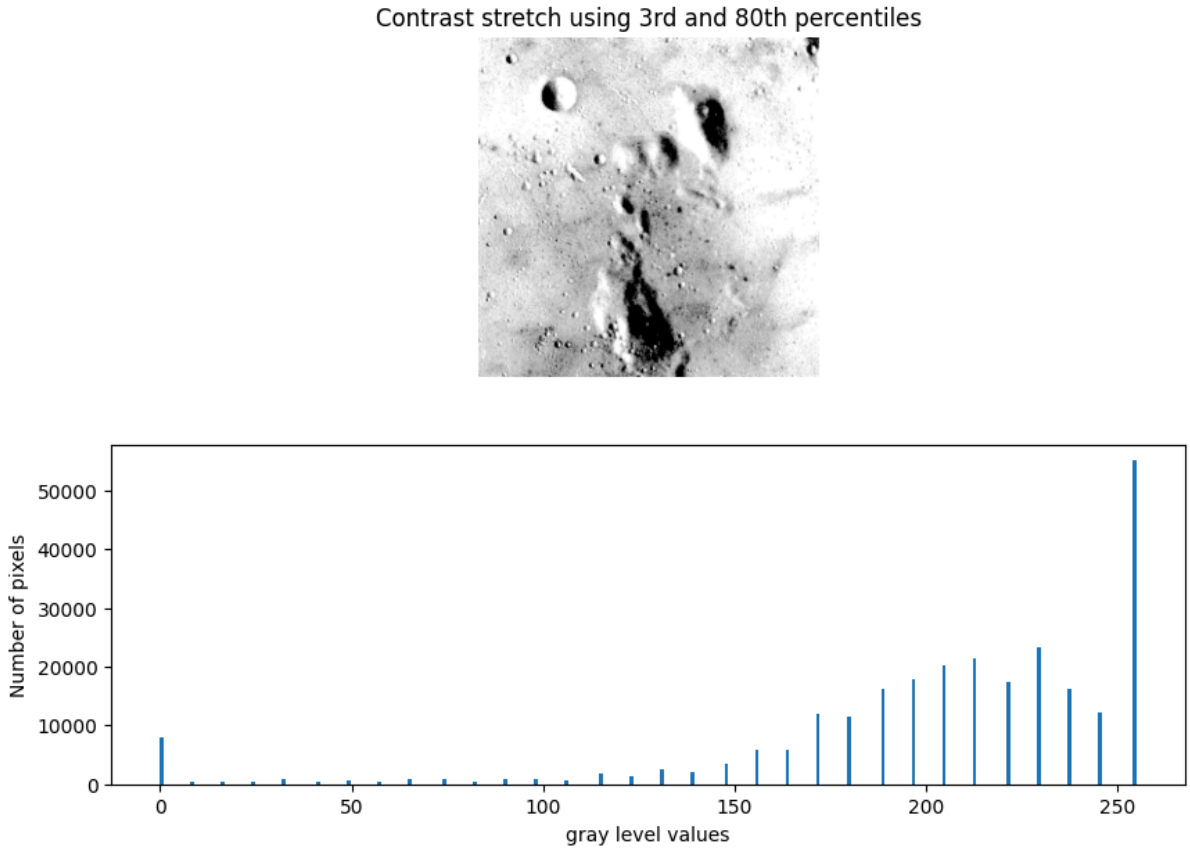
In [4]:
```
p2, p98= np.percentile(img, (2, 98))
image_rescale = exposure.rescale_intensity(img, in_range=(p2, p98))
```

In [5]:
```
fig = plt.figure(figsize=(10, 7))
fig.add_subplot(2, 1, 1)
plt.imshow(img, cmap = 'gray')
plt.axis('off')
plt.title('Original image')
fig.add_subplot(2, 1, 2)
plt.hist(img.flat, bins = 256, range=(0, 255))
plt.xlabel('gray level values')
plt.ylabel('Number of pixels')
plt.show()
fig = plt.figure(figsize=(10, 7))
fig.add_subplot(2, 1, 1)
plt.imshow(image_rescale, cmap = 'gray')
plt.axis('off')
plt.title('Contrast stretch')
fig.add_subplot(2, 1, 2)
plt.hist(image_rescale.flat, bins = 256, range=(0, 255))
plt.xlabel('gray level values')
plt.ylabel('Number of pixels')
plt.show()
```




Contrast stretch
Number of pixels
0
5000
10000
15000
20000
0
50
100
150
200
250
gray level values

In [9]:
```
# Task 1 rescaling the image using 3rd and 80th percentiles
p3,p80 = np.percentile(img, (3, 80))
img_rescale = exposure.rescale_intensity(img, in_range=(p3, p80))
plt.figure(figsize=(10, 7))
plt.subplot(2, 1, 1)
plt.imshow(img_rescale, cmap = 'gray')
plt.axis('off')
plt.title('Contrast stretch using 3rd and 80th percentiles')
plt.subplot(2, 1, 2)
plt.hist(img_rescale.flat, bins = 256, range=(0, 255))
plt.xlabel('gray level values')
plt.ylabel('Number of pixels')
plt.show()
```



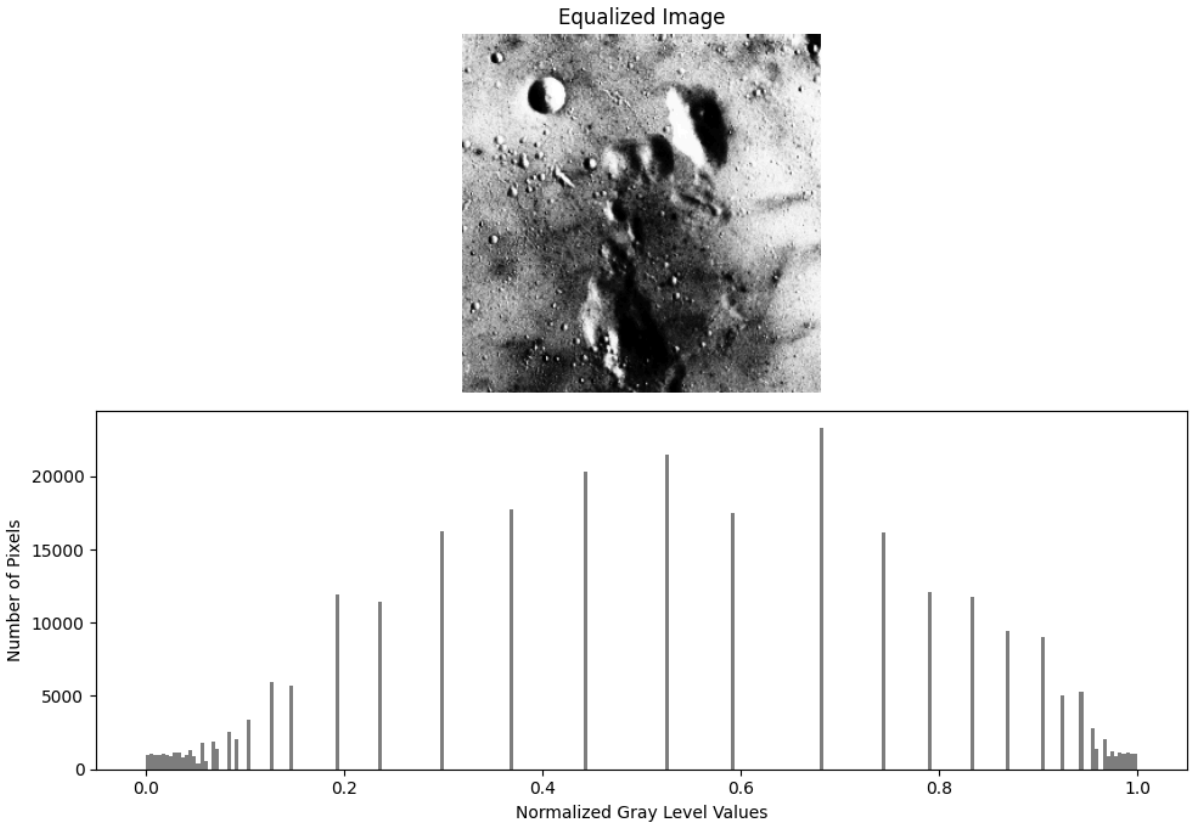
In [7]:
```
#Task 2 histogram equalization
img_eq = exposure.equalize_hist(img)

plt.figure(figsize=(10, 7))

plt.subplot(2, 1, 1)
plt.imshow(img_eq, cmap='gray')
plt.axis('off')
plt.title('Equalized Image')

plt.subplot(2, 1, 2)
plt.hist(img_eq.ravel(), bins=256, range=(0, 1), color='gray')
plt.xlabel('Normalized Gray Level Values')
plt.ylabel('Number of Pixels')

plt.tight_layout()
plt.show()
```



In [8]:
```
#Task 3 histogram matching

image = data.chelsea()
reference = data.rocket()

matched = exposure.match_histograms(image, reference, channel_axis=-1)

fig, axes = plt.subplots(nrows=1, ncols=3, figsize=(15, 5))

axes[0].imshow(image)
axes[0].set_title('Source Image (Chelsea)')
axes[0].axis('off')

axes[1].imshow(reference)
axes[1].set_title('Reference Image (Rocket)')
axes[1].axis('off')

axes[2].imshow(matched)
axes[2].set_title('Matched Image')
axes[2].axis('off')

plt.tight_layout()
plt.show()
```

